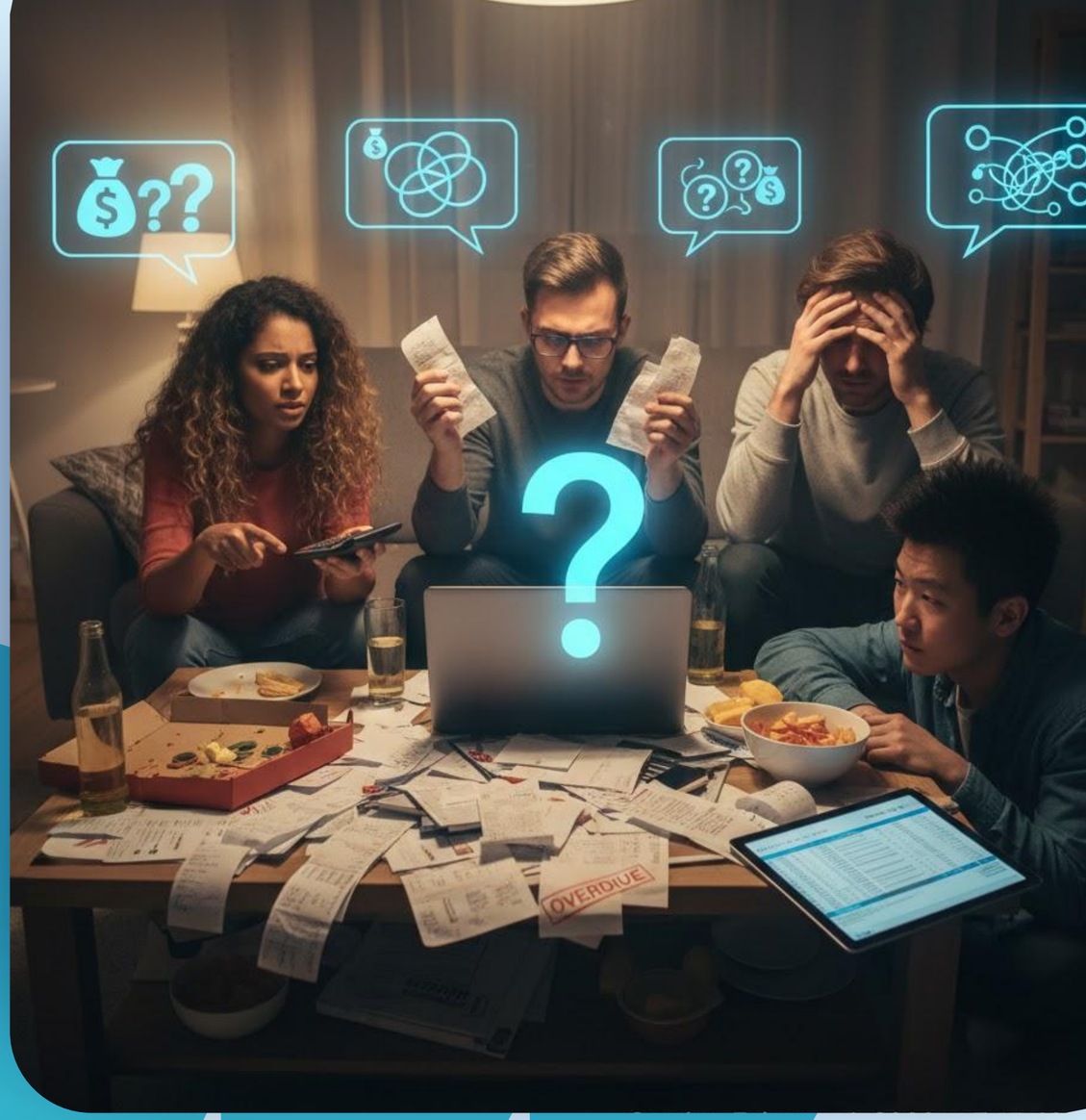
FairShare

Transparent Expenses, Hassle-Free Settlements

Sprint 0: Release Planning



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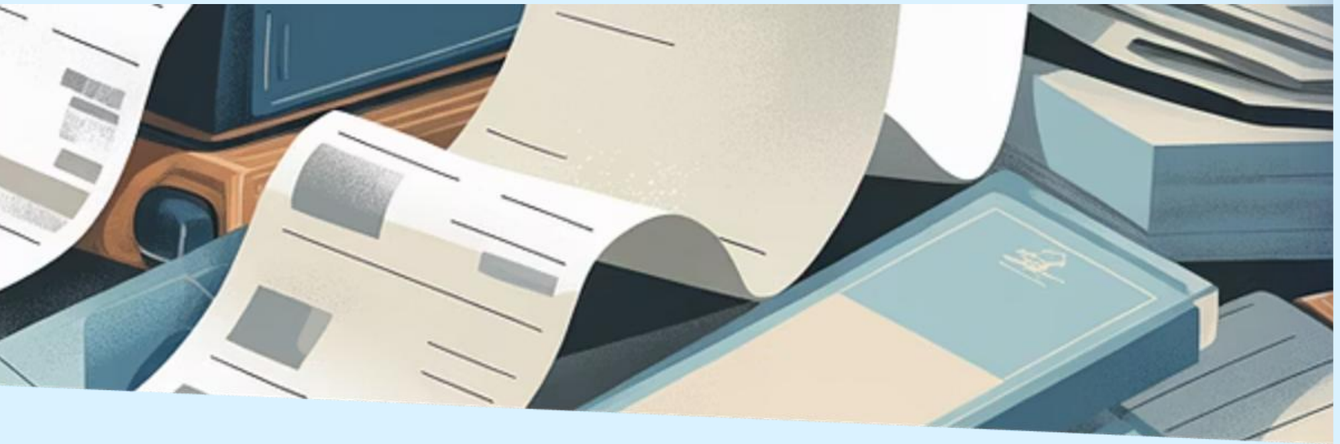
Why FairShare ?

THE PROBLEM

- Shared expenses are confusing and error-prone.
- People forget who paid what
  → conflicts and misunderstandings.
- Manual calculations are tedious and inefficient.



SOLUTION

Automatic Calculations

FairShare intelligently splits expenses and calculates balances for all members.

Transparent Records

Every transaction is permanently recorded for accountability and trust.

Smart Settlements

Minimize debt chains with automated “A pays B $X” recommendations.

Visual Dashboards

Users instantly see who owes whom, net balances, and total group spending.

Flexible Splits

Support for equal, percentage, or exact amount splits among selected members.

Name: Sarah Johnson

- Age: 27, lives with 3 roommates
- Goal: Keep track of shared rent, groceries, and utilities without confusion.
- Frustration: Manual tracking leads to mistakes and delayed repayments.

How FairShare Helps:

- Automatically calculates balances
- Provides transparent dashboard of who owes whom
- Suggests optimal settlements to clear debts quickly

USER ROLES AND PERSONA

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1 ADMIN

Description:
Manages the group, assigns roles, approves settlements.

Jobs to Get Done:

- Create groups for household/trip/team
- Invite members via code
- Assign admin/member roles
- Approve expense edits or deletions
- Review overall balances and settlements

2 MEMBERS

Description:
Participates in the group, adds expenses, and views balances.

Jobs to Get Done:

- Add/edit/delete personal or shared expenses
- Split expenses equally, by percentage, or exact amounts
- View who owes them or whom they owe
- Track group total spending
- Receive automated settlement suggestions

WHO CAN USE FAIRSHARE?

- Households sharing rent, groceries, or utilities
- Friends or family on trips sharing travel and meal expenses
- Teams collaborating on shared office or project expenses
- Flatmates, student groups, and even small clubs or communities

PRODUCT BACKLOG

EPICS AND STORIES

EPIC 1: USER & GROUP MANAGEMENT

- **Signup / Login**: As a user, I want to create an account or login so that I can securely access FairShare.
- **Create Group**: As an admin, I want to create a household/trip/team group so that members can share expenses.
- **Join Group via Invite Code**: As a member, I want to join a group using an invite code so that I can participate in shared expenses.
- **Assign Roles**: As an admin, I want to assign roles (admin/member) so that permissions are properly managed.

EPIC 2: EXPENSE MANAGEMENT

- **Add Expense**: As a member, I want to add an expense with title, amount, paid by, date, participants, and category so that all expenses are recorded.
- **Edit/Delete Expense**: As a member, I want to edit or delete an expense so that records stay accurate.
- **Attach Notes/Receipts**: As a member, I want to attach notes or upload receipts so that expense details are clear and verified.

EPIC 3: SMART SPLIT OPTIONS

- **Equal Split**: As a member, I want expenses to be split equally among participants so that the cost is fair.
- **Percentage Split**: As a member, I want to split expenses by percentage so that contributions are flexible.
- **Exact Amount Split**: As a member, I want to assign exact amounts to participants so that custom splits are possible.
- **Split Among Selected Members Only**: As a member, I want to include only certain members in the split so that irrelevant members are excluded.

EPIC 4: BALANCE DASHBOARD

- **Who Owes Whom**: As a member, I want to see who owes whom so that I know my debts or credits.
- **Net Balance Per User**: As a member, I want net balance displayed (green = gets money, red = owes money) so that finances are clear.
- **Group Total Spending**: As a member, I want to see the total group spending so that I understand overall expenses.
- **Balance Sheet Display**: As a member, I want a detailed balance sheet so that all transactions are transparent.

EPIC 5: SETTLEMENT PLAN

- **Automatic Settlement**: As a member, I want FairShare to generate “A pays B $X” automatically so that debts are minimized.
- **Minimum Transactions**: As a member, I want the system to reduce the number of transactions so that settlement is efficient.
- **Clear All Debts**: As a member, I want to mark all debts as settled once payments are made so that the ledger stays updated.

BACKLOG PRIORITIZATION

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Goal: Deliver the most valuable features first to ensure a functional product.

HIGH PRIORITY

- Signup/Login (Admin & Member access)
- Create/Join Group
- Add/Edit/Delete Expense
- Equal Split of Expenses
- Dashboard showing who owes whom & net balances

LOW PRIORITY

- Percentage/Exact amount splits
- Notes & receipt upload
- Automated debt minimization & optimized settlements
- Advanced dashboard visualizations

MINIMUM VIABLE PRODUCT (MVP)

The simplest usable version of FairShare that delivers value to users.

MVP for FairShare Includes:

- Secure signup/login for users
- Group creation and joining via invite code
- Adding, editing, and deleting expenses
- Equal split of expenses among participants
- Basic dashboard showing balances and total spending

MVP Goal:

- Ensure users can track expenses transparently and see balances clearly.
- Lay foundation for future enhancements in Sprints 2 & 3.

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SPRINT PLANNING

RELEASE PLAN – 3 SPRINTS

Based on the current FairShare product backlog, team capacity, and Sprint 1 story point estimates using the modified Fibonacci sequence, the project is estimated to require three (3) sprints to deliver a usable and demonstrable product by the end of the term.

SPRINT 1

- Signup/Login
- Create/Join Group
- Add/Edit/Delete expense
- Equal split functionality
- Balance Dashboard (basic)

SPRINT 2

- Percentage & exact amount split
- Notes & receipt upload
- Visual enhancements in dashboard
- Settlement plan calculation

SPRINT 3

- Auto debt minimization
- Final dashboard enhancements
- Testing & bug fixes
- Final release preparation

Loan Default Prediction

By monitoring payment patterns and external economic factors, AI predicts the likelihood of customers defaulting on loans.

Market Risk Analysis

AI models forecast fluctuations in currency, stocks, and commodities, helping investors and banks minimize exposure to volatile markets.

SPRINT 1 - Tasks per story

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1 EPIC – 1 USER & GROUP MANAGEMENT

Story 1: Signup / Login
Tasks:
- Design signup/login UI in React
- Create backend API endpoints (Java) for signup/login
- Connect frontend forms to backend endpoints
- Validate input (email, password rules)
- Implement password hashing & secure authentication
- Test signup/login functionality

Story 2: Create Group
Tasks:
- Design group creation form in UI
- Backend API endpoint for creating a group
- Generate unique group code for invite
- Associate group with the creating admin in DB
- Test group creation with multiple users

Story 3: Join Group via Invite Code
Tasks:
- Design "Join Group" form in UI
- Backend API to verify invite code and add member
- Update group-member relationships in DB
- Test joining group functionality

2 EPIC – 2 EXPENSE MANAGEMENT

Story 1: Add Expense
Tasks:
- Create "Add Expense" UI form (title, amount, paid by, date, participants, category)
- Backend API to add expense to DB
- Validate inputs (amount > 0, date format)
- Update balance calculations upon adding expense
- Test adding expense with different participants

Story 2: Edit/Delete Expense
Tasks:
- Implement Edit/Delete buttons in UI
- Backend API to update or delete expense
- Ensure balance recalculation after edits/deletes
- Test edit/delete with multiple users and scenarios

3 EPIC – 3 SMART SPLIT OPTIONS

Story 1: Equal Split
Tasks:
- Implement equal split logic in backend
- Connect backend logic to balance calculation
- Update UI to display split amounts per participant
- Test with various numbers of participants and expense amounts

4 EPIC – 4 BALANCE DASHBOARD

Story 1: Who Owes Whom & Net Balance
Tasks:
- Design dashboard UI showing user balances (green/red)
- Backend API to fetch balances for group
- Calculate net balance per user
- Update balances dynamically when expense is added/edited/deleted
- Test dashboard for correctness

Story 2: Group Total Spending
Tasks:
- Add total spending summary to dashboard UI
- Backend API to compute total group spending
- Display in a clear, readable format

SPRINT 1 – User Stories, estimates and acceptance criteria

Story 1: User Signup & Login
User Story:
As a user, I want to sign up and log in so that I can securely access FairShare.
Estimate: 5 Story Points
Acceptance Criteria:
User can create an account with valid credentials
User can log in using registered credentials
Invalid login attempts display an appropriate error message
User session remains active until logout
Unauthorized users cannot access protected pages

Story 2: Create Group
User Story:
As an admin, I want to create a group so that I can manage shared expenses.
Estimate: 3 Story Points
Acceptance Criteria:
Admin can create a group by providing a group name
System generates a unique invite code
Creator is automatically assigned admin role
Group is visible to the creator after creation

Story 3: Join Group via Invite Code
User Story:
As a user, I want to join a group using an invite code so that I can participate in shared expenses.
Estimate: 3 Story Points
Acceptance Criteria:
User can enter an invite code to join a group
Valid invite code adds the user to the group
Invalid invite code displays an error message
User can see the group once joined

Story 4: Add Expense
User Story:
As a member, I want to add an expense with details so that it is recorded accurately.
Estimate: 8 Story Points
Acceptance Criteria:
User can enter expense title, amount, payer, date, participants, and category
Amount must be greater than zero
Expense is saved successfully
Expense appears in the group's expense list
Balances update automatically after adding an expense

Story 5: Edit/Delete Expense
User Story:
As a member, I want to edit or delete an expense so that incorrect entries can be fixed.
Estimate: 5 Story Points
Acceptance Criteria:
User can edit existing expense details
User can delete an expense
Changes are reflected immediately
Balances are recalculated after edit or delete
Only authorized users can modify expenses

Story 6: Equal Expense Split
User Story:
As a member, I want expenses to be split equally so that costs are shared fairly.
Estimate: 5 Story Points
Acceptance Criteria:
Expense amount is divided equally among selected participants
Split values are calculated correctly
Split amounts are visible to all participants
Balances update correctly after splitting

Story 7: Basic Balance Dashboard
User Story:
As a member, I want to see who owes whom so that I understand my financial position.
Estimate: 8 Story Points
Acceptance Criteria:
Dashboard displays net balance per user
Users who owe money are shown in red
Users who are owed money are shown in green
Group total spending is displayed
Dashboard updates when expenses change

ITEMS NOT EXPECTED TO BE COMPLETED

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INITIAL DEFINITION OF DONE

Out-of-Scope / Stretch Features

- Advanced analytics and reports
- Multi-currency support
- Mobile application

Reason

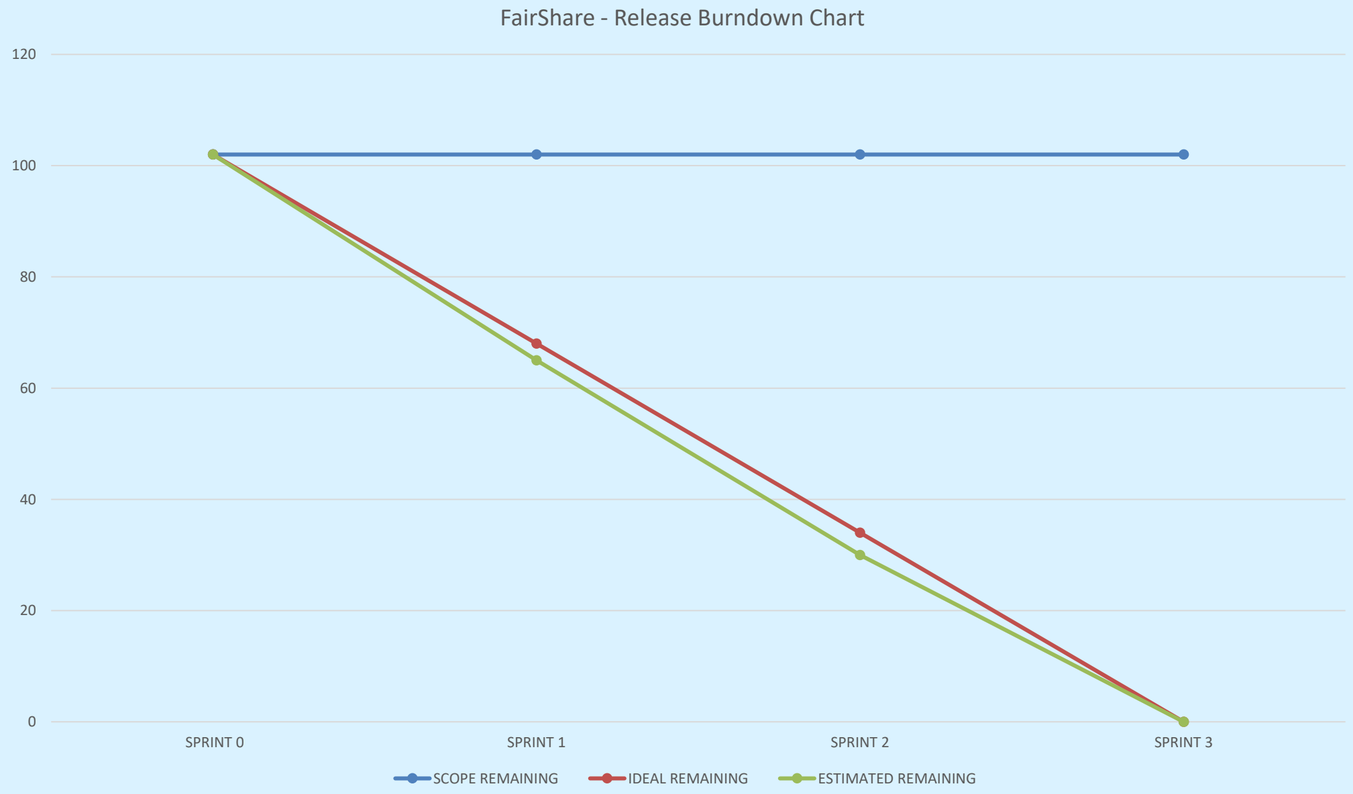
- Lower priority for MVP
- Time and scope constraints

A user story is considered Done when:

- Code is implemented and committed to GitHub
- Code compiles and runs without errors
- Acceptance criteria are met
- Unit tests pass (where applicable)
- Code is reviewed by at least one team member
- Documentation is updated if required

Initial Burndown Chart

This release burndown chart represents the entire FairShare project across all planned sprints, including those beyond the course, showing total scope, ideal progress, and estimated progress based on sprint velocity



SPRINT	SCOPE REMAINING	IDEAL REMAINING	ESTIMATED REMAINING
SPRINT 0	102	102	102
SPRINT 1	102	68	65
SPRINT 2	102	34	30
SPRINT 3	102	0	0

✅ **Sprint Length**
Each sprint = 2 weeks
Total sprints = 3
✅ **Story Points per Sprint (Planned Velocity)**
From Sprint 1 planning:
Sprint 1 velocity = 37 points
Sprint 2 = 35 points
Sprint 3 = 30 points
✅ **Total Scope**
Total release scope = 102 story points

Initial Burndown Chart

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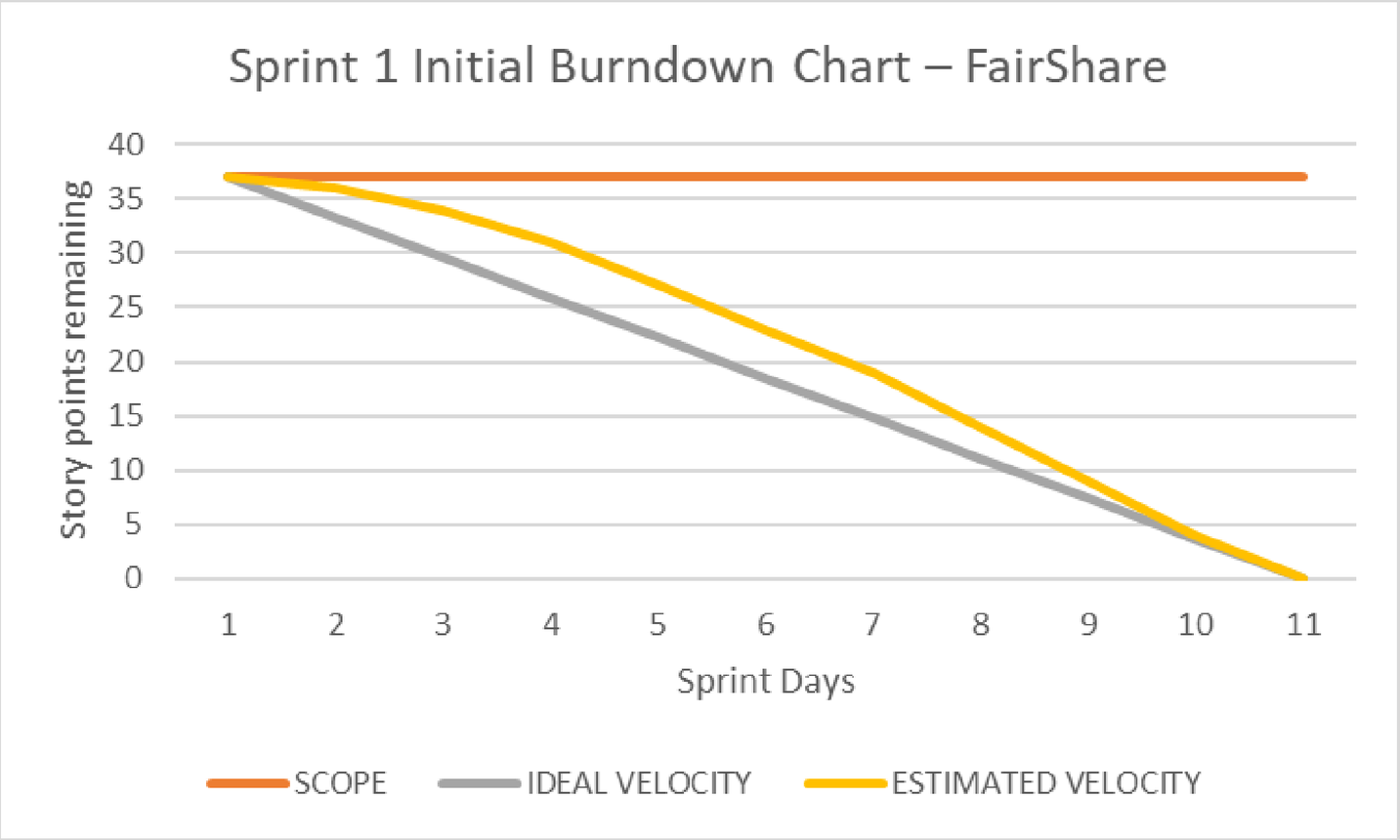
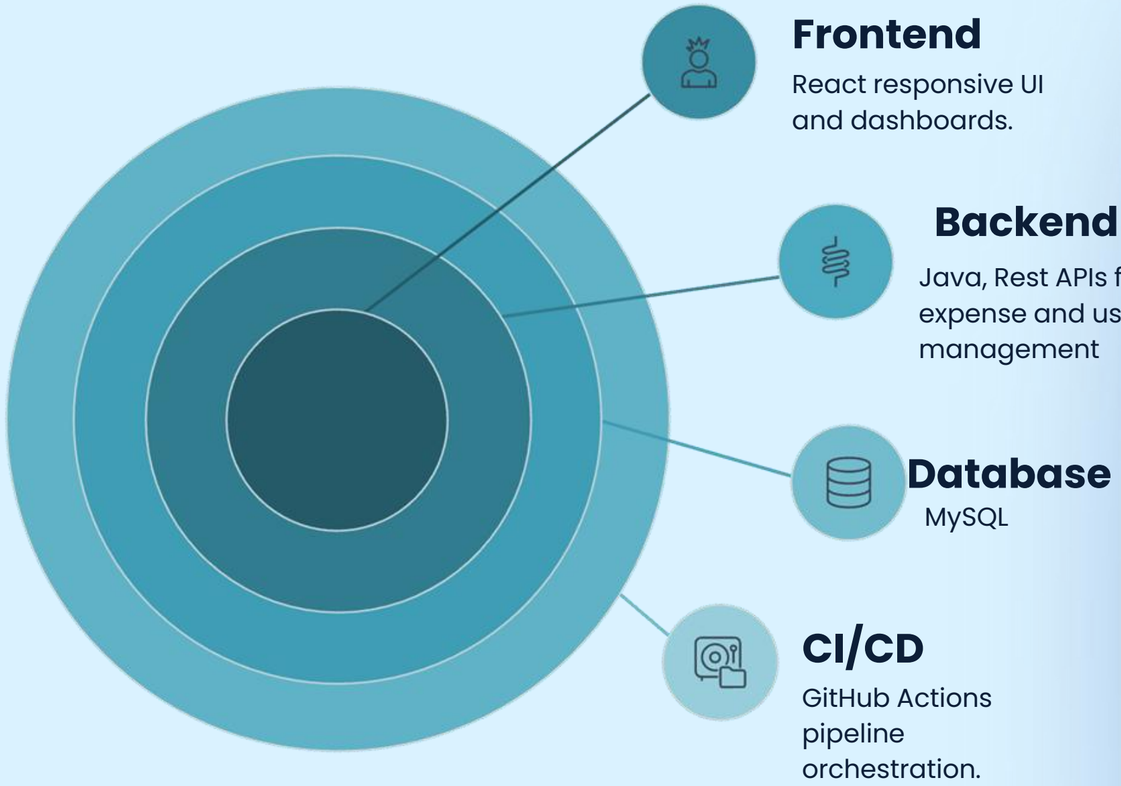
This is an initial burndown chart based on estimated story points for Sprint 1.
The scope line represents our committed work, the ideal velocity shows perfect linear progress, and the estimated velocity reflects a more realistic execution pace.



Story	Estimation (Fibonacci)
Signup / Login	5
Create Group	3
Join Group via Invite Code	3
Add Expense	8
Edit/Delete Expense	5
Equal Split	5
Basic Balance Dashboard	8

We used a modified Fibonacci sequence so that story estimates capture relative effort and uncertainty rather than exact time.



SYSTEM ARCHITECTURE

React Frontend

Responsive, intuitive user interface with real-time dashboard visualisations and interactive expense management.

Java Backend

Spring Boot REST APIs handling expense management, user authentication, and intelligent splitting algorithms.

MySQL Database

Persistent storage for transactions, user records, group data, and comprehensive audit trails.

GitHub Actions

Automated CI/CD pipeline for builds, testing, and seamless deployment to production environments.

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RETROSPECTIVE

STOP
What we should stop doing

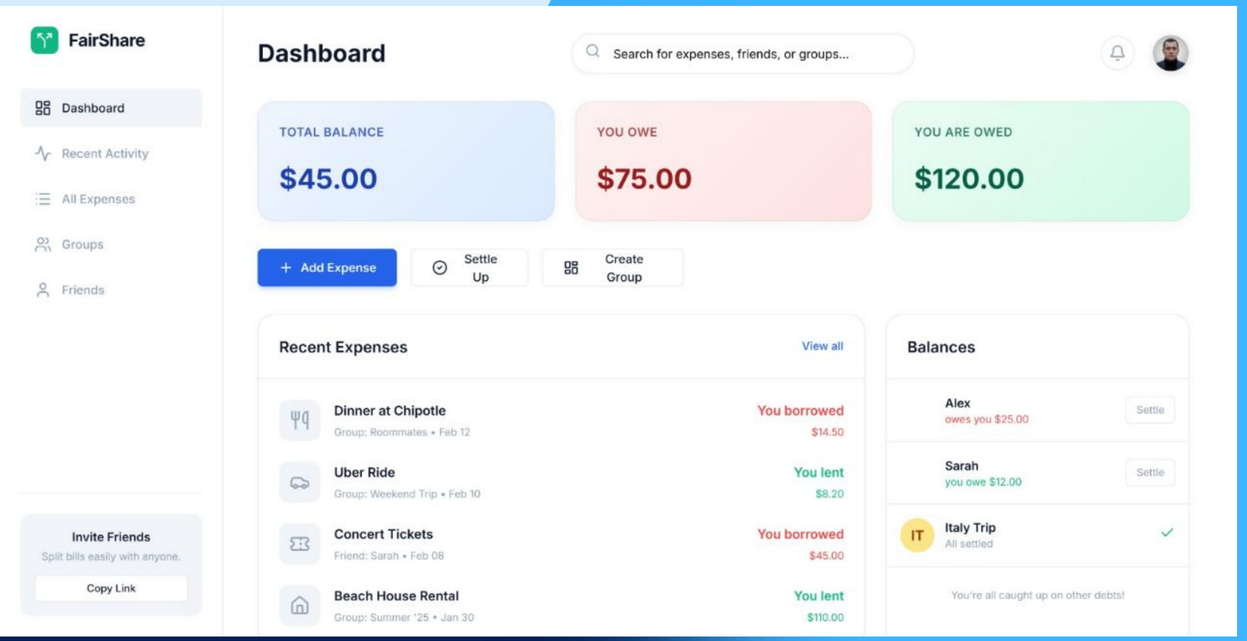
Overloading early discussions with too many future features
We realized that focusing too much on advanced features early slowed down decision-making for the MVP.
Assuming shared understanding without confirmation
Some backlog items initially meant different things to different team members.
Unstructured task ownership
Early planning lacked clear ownership for stories, which could cause confusion later.

START
What we should start doing

Clearly defining acceptance criteria before implementation
Each story should have well-defined acceptance criteria before being pulled into a sprint.
Assigning owners for every story and task
This ensures accountability and smoother sprint execution.
Using GitHub issues consistently
All stories and tasks will be tracked as GitHub issues and linked to commits.
Validating scope early
Confirm what is in scope vs. out of scope at the beginning of each sprint to avoid scope creep.

CONTINUE
What we should keep doing

Collaborative backlog refinement
Group discussions helped us prioritize the MVP effectively.
Open communication during planning
Asking questions and clarifying assumptions early improved alignment.
Breaking epics into clear user stories
This made sprint planning more manageable and realistic.
Regular team check-ins
Weekly meetings helped keep everyone aligned and informed.



Project layout Design

THANK YOU FOR YOUR ATTENTION

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